Hiten Dalmia

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Education

Chennai Mathematical Institute — Chennai, India
Master of Science in Theoretical Computer Science (On Full Scholarship) — *Aug. 2026 – Discontinued*

Kirori Mal College, University of Delhi — Delhi, India
Bachelor of Science (Honours with Research) in Mathematics, Minor in Economics — *Nov. 2022 – May 2026*

Experience

Full-Stack Architect — Mar. 2023 – Present
Freelance & Fractal — *Remote*

- Delivered 10+ production web applications for client businesses via Upwork and Fractal, specializing in full-stack architecture, automated testing, and AI-assisted development.
- **Order Procurement & Management Solution** | *Next.js, Node.js, PostgreSQL, TypeScript*
  - Architected a full-stack order portal using Next.js, ExpressJS, and TypeORM, applying Clean Architecture and Domain-Driven Design (DDD) to isolate core domain logic.
  - Designed modular React customer and administrative portals applying micro frontends principles to support real-time inventory listings, task dispatching, and recurring schedules.
  - Built an automated order ingestion pipeline using message broker queues to parse incoming order messages into 500+ structured database records daily, eliminating manual entry.
  - Structured relational PostgreSQL database schemas with CQRS query optimization and indexed constraints to process transaction spikes with sub-50ms query response times.

Software Engineer — Oct. 2023 – Apr. 2025
Tagglabs Experimental Pvt. Ltd. (Contract) — *Gurugram, Haryana (Remote)*

- **Cross-platform AR App** | *Flutter, Django, OpenCV, C* **(Live link)**
  - Engineered a native C/C++ bridge using `dart:ffi` to integrate OpenCV SIFT algorithms into Flutter, enabling cross-platform computer vision across Android and iOS from a single codebase.
  - Constructed a real-time camera feed processing pipeline normalizing OS frame buffers into C structs, achieving zero memory leaks and sub-33ms frame latency at 30+ FPS.
  - Implemented real-time keypoint feature matching, transmitting object recognition coordinates asynchronously to Dart for dynamic augmented reality UI overlay rendering.
  - Developed RESTful backend APIs in Django applying SOLID principles and built responsive Flutter UI modules to power location-aware spatial discovery features for alpha launch.
- **Kia National Conference App** | *Flutter*
  - Orchestrated event management software in Flutter for 500+ attendees at the Kia National Conference in Bangkok, featuring real-time registration, attendee directories, and interactive navigation.

Technical Skills

Architecture & Backend: Node.js, ExpressJS, NestJS, TypeScript, Clean Architecture, DDD, CQRS, PostgreSQL, Django
Frontend & Messaging: React, Next.js, Micro Frontends, Microservices, Event-Driven Messaging, RESTful APIs, Flutter
Languages & CV: Python, C++, C, SQL, Dart, OpenCV (SIFT), NumPy, Pandas, Scikit-learn, Matplotlib
Cloud, DevOps & AI Tools: AWS, Docker, Git, CI/CD, Linux, Cursor, Copilot, LLM Pipeline & Context Engineering

Achievements

Erasmus Mundus Joint Master MATHS-DISC Fellowship — Europe
Issued by European Union and affiliated universities — *Mar. 2026*

Master in Mathematics (ISI) and MSc Computer Science (CMI) Scholarships — India
Indian Statistical Institute and Chennai Mathematical Institute respectively — *Jun. 2026*

Publications

Computable Analysis: Mathematical Foundations and Computability of Real Functions
Poincare Journal of Mathematical Physics & Applications, Vol. 1, No. 1 (2026), 1–20 (with Sujit Kumar)